Software Requirements Specification
for
LendWise – MSME Financial Risk Management System

Version 1.0 approved

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Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document specifies the software requirements for **LendWise – MSME Financial Risk Management System**, Version 1.0. LendWise is a banking software solution proposed for PrimeLend to support MSME credit risk assessment, fraud detection, analytics, and explainable AI–based decision support. This SRS describes the complete system and all its major functional and non-functional requirements.

1.2 Document Conventions

This document follows IEEE Software Requirements Specification standards. The word **"shall"** is used to indicate mandatory requirements, **"should"** indicates recommended requirements, and **"may"** indicates optional requirements. Section numbering and formatting follow standard IEEE conventions. All requirements are stated clearly and uniquely to avoid ambiguity.

1.3 Intended Audience and Reading Suggestions

This document is intended for bank stakeholders, project managers, developers, testers, risk analysts, and compliance teams. Readers should begin with Sections 1 and 2 to understand the system scope and context. Developers and testers should focus on Sections 3 and 4 for detailed system features and interface requirements, while managers and stakeholders may focus on scope, objectives, and non-functional requirements.

1.4 Product Scope

LendWise is an integrated financial risk management software designed to improve MSME lending decisions at PrimeLend. The system uses machine learning–based credit scoring, rule-based fraud detection, analytics dashboards, and an explainable AI chatbot to enhance decision accuracy, reduce fraud risk, improve transparency, and ensure regulatory compliance. The product supports PrimeLend's business strategy of responsible and data-driven lending.

1.5 References

1. IEEE 29148 – Software Requirements Specification Standard
2. PrimeLend Internal Risk Management and Lending Policies
3. Financial Regulatory Compliance Guidelines
4. Machine Learning Best Practices for Financial Service.

2. Overall Description

2.1 Product Perspective

LendWise is a new, self-contained financial risk management software developed for PrimeLend to support MSME lending operations. The system is designed to integrate with existing core banking systems and transaction databases. It does not replace current banking systems but enhances them by providing advanced credit risk assessment, fraud detection, analytics, and explainable AI capabilities. LendWise functions as a decision-support layer within the larger banking ecosystem and interfaces with data sources such as customer profiles, transaction histories, and loan management systems.

2.2 Product Functions

At a high level, LendWise provides the following major functions:

- Evaluate MSME creditworthiness using machine learning models
- Detect and flag potentially fraudulent transactions
- Provide dashboards and analytical reports for risk monitoring
- Explain credit and risk decisions using an AI-based chatbot
- Support role-based access for different user groups

2.3 User Classes and Characteristics

The following user classes are expected to use the system:

- **Loan Officers:** Frequent users responsible for reviewing loan applications and credit scores. Moderate technical expertise.
- **Risk Analysts:** Advanced users who monitor portfolio risk and fraud alerts. High analytical expertise.
- **Bank Managers:** Occasional users who view dashboards and summary reports. Limited technical expertise.
- **Auditors and Compliance Officers:** Periodic users who review decision explanations and logs for regulatory compliance. High domain knowledge with limited system interaction.

Loan Officers and Risk Analysts are the primary users of the system.

2.4 Operating Environment

LendWise will operate in a secure banking environment and will be accessible through modern web browsers. The system will run on bank-approved servers and will integrate with existing databases and backend services. It must coexist with PrimeLend's core banking applications without performance interference.

2.5 Design and Implementation Constraints

The system must comply with banking and financial regulatory policies. Secure data handling and encryption are mandatory. Integration with existing banking systems and databases is required. The use of explainable machine learning models is necessary to meet regulatory transparency requirements. Development must follow PrimeLend's security, coding, and deployment standards**.**

2.6 User Documentation

The following user documentation will be provided with the software:

- User manual for loan officers and analysts
- Administrator guide
- Online help and usage instructions
  All documentation will be delivered in digital format

2.7 Assumptions and Dependencies

It is assumed that historical MSME financial and transactional data will be available for analysis. The system depends on successful integration with PrimeLend's core banking systems and databases. The availability of secure network infrastructure and required APIs is also assumed. Any changes in regulatory requirements or data availability may impact system functionality

3. External Interface Requirements

3.1 User Interfaces

LendWise provides a web-based graphical user interface for all user interactions. The interface includes dashboards for credit risk and fraud monitoring, data tables, charts, alerts, and an AI-based chatbot window. Standard user interface elements such as navigation menus, action buttons, help links, and error messages appear consistently across all pages. Interfaces are designed to support role-based access and present information according to user privileges. Detailed screen designs are documented separately.

3.2 Hardware Interfaces

The system interfaces with standard computing hardware such as desktop and laptop computers used within the bank's network. No specialized hardware interfaces are required. All hardware interactions occur through standard input and output devices such as keyboard, mouse, and display.

3.3 Software Interfaces

LendWise interfaces with the following software components:

- Core banking systems for customer and loan data
- Transaction databases for financial and behavioral data
- Database management systems for data storage

- Machine learning services for credit scoring
- Analytics and reporting tools

Data exchanged includes loan application data, transaction records, credit scores, fraud alerts, and audit logs. Communication between software components occurs through secure APIs. Shared data must follow bank-approved data access and security standards.

3.4 Communications Interfaces

The system uses secure network communication protocols such as HTTP and HTTPS for data exchange between client browsers, servers, and integrated systems. All sensitive data transmissions must be encrypted. Authentication tokens and access controls are used to secure communication channels. Data synchronization occurs in near real time based on transaction updates.

4. System Features

4.1 Credit Scoring System

4.1.1 Description and Priority

The Credit Scoring System is a core feature of the LendWise platform. It evaluates the creditworthiness of MSME customers by analyzing financial, behavioral, and transactional data using machine learning models. This feature plays a critical role in loan approval and risk assessment processes. Due to its direct impact on lending decisions and risk mitigation, it is classified as a **High priority** feature.

4.1.2 Stimulus/Response Sequences

The typical interaction sequence for the Credit Scoring System is as follows:

- The loan officer submits an MSME loan application through the system.
- The system retrieves relevant customer, financial, and transaction data from integrated sources.
- The retrieved data is validated and processed by the credit scoring engine.
- A credit score along with a risk category (e.g., low, medium, high risk) is generated.
- The system displays the credit score and risk assessment results to the user.

4.1.3 Functional Requirements

The Credit Scoring System shall support the following functional requirements:

- **REQ-CS-1:** The system shall calculate a credit score for each MSME loan application using predefined models.
- **REQ-CS-2:** The system shall classify MSME customers into risk categories based on scoring thresholds.

- **REQ-CS-3:** The system shall store all credit evaluation results securely for audit and compliance purposes.
- **REQ-CS-4:** The system shall validate input data and notify the user in case of incomplete or invalid information.

4.2 Fraud Detection System

4.2.1 Description and Priority

The Fraud Detection System monitors MSME transaction activities to identify suspicious or abnormal patterns using rule-based logic. This feature helps minimize financial losses and protects the bank from fraudulent activities. Due to its importance in risk prevention and regulatory compliance, it is considered a **High priority** feature.

4.2.2 Stimulus/Response Sequences

The fraud detection workflow includes the following steps:

- Transaction data is continuously received by the system from banking sources.
- The system analyzes transaction patterns using predefined fraud detection rules.
- Suspicious or high-risk activities are identified.
- Fraud alerts are generated and made visible to authorized users.

4.2.3 Functional Requirements

The Fraud Detection System shall meet the following requirements:

- **REQ-FD-1:** The system shall monitor MSME transaction activity in near real time.
- **REQ-FD-2:** The system shall flag suspicious transactions based on configured rules.
- **REQ-FD-3:** The system shall log all detected fraud alerts for investigation and audit.
- **REQ-FD-4:** The system shall allow authorized users to update and manage fraud detection rules.

4.3 Dashboard and Analytics

4.3.1 Description and Priority

The Dashboard and Analytics feature provides a consolidated view of credit risk metrics and fraud trends through visual representations such as charts and reports. This feature supports informed decision-making by management and risk teams. It is categorized as a **Medium priority** feature.

4.3.2 Stimulus/Response Sequences

The interaction flow for dashboards includes:

- The user logs into the LendWise system.
- The system retrieves relevant analytical data.

- Dashboards displaying charts, metrics, and summaries are presented to the user.

4.3.3 Functional Requirements

The Dashboard and Analytics module shall provide the following capabilities:

- **REQ-DA-1:** The system shall display summarized credit risk information for MSME portfolios.
- **REQ-DA-2:** The system shall visualize fraud trends and alert statistics.
- **REQ-DA-3:** The system shall allow users to filter dashboard data by time period and risk level.

4.4 Explainable AI Chatbot

4.4.1 Description and Priority

The Explainable AI Chatbot enhances transparency by providing clear, human-readable explanations for credit scoring and fraud detection decisions. This feature supports regulatory compliance and improves user understanding of system outcomes. It is assigned a **Medium priority** feature.

4.4.2 Stimulus/Response Sequences

The chatbot interaction process includes:

- The user submits a question through the chatbot interface.
- The system interprets and processes the query.
- A contextual and understandable explanation is generated and displayed.

4.4.3 Functional Requirements

The Explainable AI Chatbot shall fulfill the following requirements:

- **REQ-XAI-1:** The system shall provide clear explanations for credit scoring decisions.
- **REQ-XAI-2:** The system shall explain fraud alerts in simple and understandable language.
- **REQ-XAI-3:** The system shall respond only to queries from authorized users.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The performance requirements for the LendWise system are as follows:

- Credit scoring results shall be generated within acceptable response times to support timely loan decisions.

- At least **95% of user requests shall be processed within two seconds** under normal operating conditions.
- Fraud detection alerts shall be generated in **near real time** upon receipt of transaction data.
- Dashboard data shall refresh at predefined intervals without negatively impacting system performance.
- The system shall continue to perform efficiently during peak usage periods.

5.2 Safety Requirements

The system shall include safety measures to prevent operational and data-related risks:

- Prevent loss, corruption, or unauthorized modification of financial and customer data.
- Provide safeguards such as data validation, backup, and recovery mechanisms.
- Prevent actions that could lead to incorrect credit decisions due to incomplete or invalid data.
- Ensure all operations comply with applicable banking safety policies and operational guidelines.

5.3 Security Requirements

The security requirements of the system include:

- Enforce secure user authentication and role-based authorization.
- Allow access to system features and data only for authenticated and authorized users.
- Encrypt sensitive data during transmission and while stored in databases.
- Comply with applicable banking security standards and data privacy regulations.
- Maintain audit logs for all critical user actions and system events.

5.4 Software Quality Attributes

The following quality attributes are critical for the LendWise system:

- **Reliability:** Ensure consistent and dependable system operation.
- **Availability:** Maintain high system uptime for uninterrupted business use.
- **Scalability:** Support growth in MSME customers, transaction volume, and data size.
- **Maintainability:** Allow easy updates to models, rules, and system features.
- **Usability:** Provide an intuitive interface for non-technical users while maintaining robustness.

5.5 Business Rules

Only authorized roles such as loan officers, risk analysts, managers, and auditors shall perform specific system functions according to assigned privileges. Credit and fraud decisions shall follow PrimeLend's internal policies and regulatory requirements. All decisions and alerts shall be recorded and retained for audit and compliance purposes.

6. Other Requirements

Additional system requirements include:

- Store and manage large volumes of MSME customer, transaction, and risk-related data in a secure database.
- Support database backup, recovery, and audit logging mechanisms.
- Comply with all applicable banking laws, financial regulations, and data privacy requirements.
- Enable reuse of system components such as machine learning models and analytics modules.
- Consider support for internationalization and localization in future releases.